

WEIL POSITIVITY ON THE WINDOW OF WIDTH $\log 15$

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ABSTRACT. Weil's criterion states that the Riemann Hypothesis holds if and only if Weil's quadratic form Q is nonnegative on all admissible test functions. Restricting the support to a fixed interval $[-L, L]$ gives a weaker, unconditional question. We prove that $Q(f) \geq 2^{-2904} \|f\|_2^2$ for every complex C^2 function f supported in $[-\frac{1}{2} \log 15, \frac{1}{2} \log 15]$. The previous largest window we know of is $\frac{1}{2} \log 12$ (our previous paper). On the new window the prime 13 enters. The proof uses the bounded-comparison framework of Liu and the general tail estimate and prolate defect bound of our previous paper, with a new weight in the Schur bound for the prime operator, and quadrature estimates that do not assume $L < 5/4$. Positivity is reduced to a finite matrix inequality of size 832 in each parity, which we certify in Arb ball arithmetic with all quadrature, rounding and inversion errors paid. The certified finite margin is about 2.40×10^{-70} (even) and 3.95×10^{-66} (odd), while the total error is below 5×10^{-94} . This is an external certificate, not a formal proof. The result concerns one fixed window and does not address the Riemann Hypothesis.

Zenodo, doi:10.5281/zenodo.23281236. Certificate code: release v1.6.0 of <https://github.com/moriryota/prolate-weil-bounds>, doi:10.5281/zenodo.23092422; data: <https://github.com/moriryota/weil-log15-certificate-data>, doi:10.5281/zenodo.23281176. Licence: CC BY 4.0.

Status. Analytic proof plus an external interval-arithmetic certificate (Arb/FLINT). The analytic error budget and the certificate passed independent AI reviews, including an independent reimplementations of the final check; no blocking issues remained. Not formalized in Lean. Not yet reviewed by independent human experts.

1. INTRODUCTION

1.1. Background. Let Q be Weil's quadratic form [10, 2]. Weil's criterion states that RH is equivalent to $Q(f) \geq 0$ for all admissible f . If the support of f is restricted to $[-L, L]$, only prime powers $n < e^{2L}$ contribute and the question can be decided unconditionally for small L . Positivity is known for $L = \frac{1}{2} \log 2$ (Yoshida [11]), $L = 0.8$ (Zhu [12]), $L = \frac{1}{2} \log 5$ (Lean 4, [6]), $L = 17/16$ (Liu [5], unrefered), $L = \frac{1}{2} \log 10$ ([8]) and $L = \frac{1}{2} \log 12$ ([9]). For $\mu = e^{2L}$, the bottom $\lambda_{\min}$ of the spectrum of the restricted form satisfies $\lambda_{\min} \leq 3.50 \times 10^{24} \mu^{9/2} \log \mu e^{-4\pi\mu}$ for $50 \leq \mu \leq 10^4$ [7, Theorem K], so each enlargement of the window must resolve a much smaller margin.

In [9] we proved a lower bound for the defect of the top prolate eigenvalue (Theorem N there), which gives an out-of-band estimate for every L , and we wrote the bounded comparison with all constants as functions of L , Ω , m and the truncation order N . That paper noted that the obstacle for larger windows is the prime operator: as e^{2L} grows, its norm grows, and with it the band limit Ω , the order N and the precision. This paper carries the method to $e^{2L} = 15$.

1.2. Result. For $L > 0$, $f \in C^2(\mathbb{R}; \mathbb{C})$ with $\text{supp } f \subset [-L, L]$, put $F_f(z) = \int f(x) e^{izx} dx$, $a(t) = \text{Re } \psi(\frac{1}{4} + \frac{it}{2}) - \log \pi$ and

$$(1.1) \quad \begin{aligned} Q(f) = & 2 \left| \int f(x) \cosh \frac{x}{2} dx \right|^2 - 2 \left| \int f(x) \sinh \frac{x}{2} dx \right|^2 + \frac{1}{2\pi} \int_{\mathbb{R}} a(t) |F_f(t)|^2 dt \\ & - 2 \sum_{n < e^{2L}} \frac{\Lambda(n)}{\sqrt{n}} \text{Re} \int_{\mathbb{R}} f(x) \overline{f(x - \log n)} dx. \end{aligned}$$

By the Guinand–Weil explicit formula, $Q(f) = \operatorname{Re} \sum_{\rho} m_{\rho} F_f(\gamma_{\rho}) \overline{F_f(\overline{\gamma_{\rho}})}$ with $\gamma_{\rho} = (\rho - \frac{1}{2})/i$; RH is not assumed. The derivation, the convergence of the zero sum, and the reduction to real symmetric operators on each parity are as in [8, Section 2]; nothing there depends on L . For $L = \frac{1}{2} \log 15$ the prime powers are $n \in \{2, 3, 4, 5, 7, 8, 9, 11, 13\}$.

Theorem O. *For every $f \in C^2(\mathbb{R}; \mathbb{C})$ with $\operatorname{supp} f \subset [-\frac{1}{2} \log 15, \frac{1}{2} \log 15]$,*

$$Q(f) \geq 2^{-2904} \int_{\mathbb{R}} |f(x)|^2 dx.$$

Theorem O is proved by an ordinary analytic argument together with an external certificate in ball arithmetic (Arb/FLINT [1]), produced and checked by separate programs. It is not a formal proof.

reference	half-width L	e^{2L}	method
Yoshida [11]	$\frac{1}{2} \log 2 \approx 0.347$	2	analytic + computer-assisted
Zhu [12]	0.8	4.95	interval certificate
[6]	$\frac{1}{2} \log 5 \approx 0.805$	5	Lean 4 (kernel-checked)
Liu [5]	$17/16 = 1.0625$	8.37	interval certificate (unrefereed)
[8]	$\frac{1}{2} \log 10 \approx 1.151$	10	interval certificate
[9]	$\frac{1}{2} \log 12 \approx 1.242$	12	interval certificate
Theorem O	$\frac{1}{2} \log 15 \approx 1.354$	15	interval certificate

TABLE 1. Unconditional positivity on windows $[-L, L]$. The approach of Connes and Consani [3] works under vanishing conditions on the Fourier transform and is not listed.

1.3. What changes from log 12.

- The prime 13 enters. With the weight $w(x) = 1 + \frac{2}{3}(x/L)^2$ in the Schur test we obtain $\|C_p\| \leq \frac{66}{16}$ (Section 3); a smaller s allows a smaller band limit Ω and hence a smaller N .
- The quadrature lemma of [9] used $L < 5/4$. Lemma 4.1 below is proved without it, with panels of length at most 64 up to $\Omega = 736$.
- The finite condition is certified at $N = 1664$ (832 modes in each parity). At $N = 1280$ the computed comparison matrix was numerically negative in both parities (Remark 3.1). The source matrices were computed one parity at a time to fit in 16 GB of memory.

Nothing else is new. In particular the tail inequality, the projection lemma and the block elimination are used exactly as stated in [9].

2. THE BOUNDED COMPARISON

We recall the reduction of [9, Section 3]. Let $L > 0$, $I = [-L, L]$, $H = L^2(I; \mathbb{C})$. For $a \in \mathbb{R}$ let $S_a f(x) = \mathbf{1}_I(x-a)f(x-a)$ and $C_p = \sum_{n < e^{2L}} \Lambda(n) n^{-1/2} (S_{\log n} + S_{-\log n})$, so that the prime term of (1.1) is $-\langle f, C_p f \rangle$. For a positive even weight w , the weighted Schur test gives $\|C_p\| \leq \operatorname{ess\,sup} R_w$, where

$$R_w(x) = \sum_{n < e^{2L}} \frac{\Lambda(n)}{\sqrt{n}} \frac{\mathbf{1}_I(x + \log n) w(x + \log n) + \mathbf{1}_I(x - \log n) w(x - \log n)}{w(x)}.$$

If $\|C_p\| \leq s$, put $m > 0$, $\gamma = s + m$, $M = \gamma I - C_p$ and $b = 2s + m$, so that $mI \preceq M \preceq bI$.

Let B_{band} have kernel $\sin(\Omega(x-y))/(\pi(x-y))$ on I , $c = \Omega L$, $d(c) = e^{-2(c+1)}/(4(c+1)(c+4))$, and

$$C = \log \frac{\Omega}{2\pi} - \frac{1}{\Omega} - \gamma, \quad E_0 = \frac{c+1}{\pi c^2}, \quad E_1 = \frac{3c^2 + 6c + 2}{\pi c^3}.$$

If $C > 0$, $c > 2$, $0 < \delta \leq d(c)$, and $u_j \geq 0$ with $u_j E_j \leq C$, then by [9, Proposition 3.1 and (3.2)]

$$(2.1) \quad Q(f) \geq C\delta \|f\|^2 + \langle f, (M + V)f \rangle, \quad V = K + U,$$

where K is the integral operator with kernel

$$K(x, y) = 2 \cosh \frac{x - y}{2} + \frac{1}{2\pi} \int_{-\Omega}^{\Omega} (a(t) - \gamma) e^{it(x-y)} dt$$

(the pole terms and the band-limited part of the archimedean term) and $U = u_0|h_0\rangle\langle h_0| + u_1|h_1\rangle\langle h_1|$ with $h_j = (I - B_{\text{band}} - \delta I)v_j$, $v_0 = (2L)^{-1/2}$, $v_1 = \sqrt{3/(2L)} x/L$.

Let $N \geq 2$, $v_j(x) = \sqrt{(2j+1)/(2L)} P_j(x/L)$, $E : \mathbb{C}^N \rightarrow H$, $Ez = \sum_{j < N} z_j v_j$, $P = EE^*$, and

$$A = E^*ME, \quad B = E^*M^2E, \quad J = E^*VE, \quad G = E^*M^{-1}E, \quad D_M = B - A^2.$$

For $\rho > 1$, Lemma 3.2 of [9] gives explicit e, n, h such that the residual $R = V - EJE^*$ satisfies $\|(I - P)RP\| \leq e$, $(I - P)R(I - P) \succeq -n$, $\|(I - P)R(I - P)\| \leq h$, and with

$$\alpha = e + \frac{2e^2}{m - n}, \quad \beta = \frac{e + n}{m^2} + \frac{2h^2}{m^2(m - n)}$$

the block elimination of [9, (3.3)] gives: if $n < m$ and $G^{-1} + J \succeq \alpha I + \beta D_M$, then $M + V \succeq 0$ on H . Finally, for every matrix X , $\mathcal{W}(X) = X + X^* - X^*AX + m^{-1}(I - AX - X^*A + X^*BX) \succeq G$, so it suffices to certify, in each parity,

$$(2.2) \quad T = \mathcal{W}(X)^{-1} + J - \alpha I - \beta(B - A^2) \succ 0.$$

3. THE WINDOW $\frac{1}{2} \log 15$

Let $L = \frac{1}{2} \log 15 \approx 1.354025$, so $e^{2L} = 15$ and the prime powers are 2, 3, 4, 5, 7, 8, 9, 11, 13.

Prime operator. Take $w(x) = 1 + \frac{2}{3}(x/L)^2$. In the variable $y = e^{2x}$, $x \geq 0$, the shift by $\pm \log n$ is switched on or off at $y = \max(15/n^2, n^2/15)$, so the breakpoints are

$$y \in \left\{ 1, \frac{16}{15}, \frac{5}{3}, \frac{49}{15}, \frac{15}{4}, \frac{64}{15}, \frac{27}{5}, \frac{121}{15}, \frac{169}{15}, 15 \right\}.$$

Bounding R_w in ball arithmetic on $9 \cdot 256 = 2304$ closed subintervals gives $R_w < 4.0915$, the maximum occurring on the cell $\frac{49}{15} \leq y \leq \frac{15}{4}$. Hence $\|C_p\| \leq s = \frac{66}{16}$. We take $m = \frac{1}{2}$, so $\gamma = \frac{37}{8}$ and $b = \frac{35}{4}$. The weight enters only the proof of the norm bound; the operator C_p and the matrices A , B do not depend on it.

Tail. With $\Omega = 736$ we get $c = \Omega L \approx 996.56$ and $C \approx 0.136994 > \frac{1}{8}$. Moreover $-\log_2 d(c) = 2900.29 \dots$, so we take $\delta = 2^{-2901}$ and $\tau = \delta/8 = 2^{-2904} < C\delta$. The coefficients $u_0 = 428$, $u_1 = 142$ satisfy $u_0 E_0 \approx 0.13684 < C$ and $u_1 E_1 \approx 0.13634 < C$.

Projection errors. With $N = 1664$ (that is, 832 modes in each parity) and $\rho = 3$, Lemma 3.2 of [9] gives, in ball arithmetic,

$$e < 2.75 \times 10^{-213} < 2^{-706}, \quad n < 1.09 \times 10^{-429} < 2^{-1424}, \quad h < 6.29 \times 10^{-428} < 2^{-1419}.$$

Then $n < m$, and $\alpha + \beta b^2 < 2^{-697}$ (checked with exact rationals).

Remark 3.1. At $N = 1280$ the LDL^* factorization of the midpoint comparison matrix had one negative pivot in each parity (smallest pivots -3.46 even and -93.5 odd). The same happened for $e^{2L} = 12$ at $N = 640$, where the ratio N/c was about 1.24; there the condition held at $N/c \approx 1.61$. Here $N/c \approx 1.28$ at $N = 1280$ and $N/c \approx 1.67$ at $N = 1664$.

4. SOURCE MATRICES, QUADRATURE AND THE CERTIFICATE

A and B. After splitting I at the support boundaries of the shifts, the entries are integrals of polynomials of degree at most 3326; Gauss–Legendre quadrature with 1666 nodes is exact, so only rounding errors occur. These are enclosed by the balls, and $\epsilon_A = \epsilon_B = 2^{-400}$ covers them together with the distance to the rational centres.

J. In each parity the band and gamma-weighted integrals are computed on the 19 panels with endpoints

$$0, \frac{1}{2}, 1, 2, 4, 8, 16, 32, 64, 128, 192, 256, \dots, 704, 736$$

(steps of 64 from 64 to 704), with 224 Gauss nodes each. For each parity $p \in \{0, 1\}$ and $j \equiv p \pmod{2}$ put

$$b_j(t) = i^{-p} F_{v_j}(t) = (-1)^{(j-p)/2} \sqrt{2L(2j+1)} j_j(Lt),$$

where j_j is the spherical Bessel function. These functions are real for real t , entire, and $|b_j(z)| \leq \sqrt{2L} e^{L|\operatorname{Im} z|}$. In a fixed parity the band and gamma-weighted band entries are $\pi^{-1} \int_0^\Omega b_i(t) b_j(t) dt$ and $\pi^{-1} \int_0^\Omega (a(t) - \gamma) b_i(t) b_j(t) dt$.

Lemma 4.1 (quadrature). *On the Bernstein ellipses $\rho = 2$ of the panels, $|\operatorname{Im} z| \leq 24$ and $|z| \leq 768$. There $|b_i b_j| \leq H_B := 2Le^{48L}$, and $g(z) = \frac{1}{2}[\psi(\frac{1}{4} + \frac{iz}{2}) + \psi(\frac{1}{4} - \frac{iz}{2})] - \log \pi - \frac{37}{8}$ is holomorphic with $|g| < 1100$. Consequently the quadrature error of J , including the pole and U terms, is below 2^{-319} in operator norm.*

Proof. For a panel of length ℓ , the ellipse has semi-axes $\frac{5}{8}\ell$ and $\frac{3}{8}\ell$. The longest panels have length 64, so $|\operatorname{Im} z| \leq 24$, and every point satisfies $|z| \leq 736 + 32$. For $w = \frac{1}{4} \pm \frac{iz}{2}$: on the three panels of length at most 1, $\operatorname{Re} w \geq \frac{1}{16}$; on the others $\operatorname{Re} z \geq \frac{7}{4}$, so $|\operatorname{Im} w| \geq \frac{7}{8}$. Hence $|w + k| \geq \frac{1}{16}$ for all integers $k \geq 0$, and g has no poles.

Since $\operatorname{Re} w \geq \frac{1}{4} - 12 = -\frac{47}{4}$, the point $v = w + 13$ has $\operatorname{Re} v \geq \frac{5}{4}$ and $|v - 1| \leq 397$. The series $\psi(v) = -\gamma_E + (v-1) \sum_{k \geq 0} \frac{1}{(k+1)(k+v)}$ [4, 5.7.6], with $|k+v| \geq k+1$, $\gamma_E < 1$ and $\sum_{k \geq 0} (k+1)^{-2} < 2$, gives $|\psi(v)| < 1 + 397 \cdot 2 = 795$. The 13 steps of the recurrence $\psi(w) = \psi(w+1) - 1/w$ cost at most $13 \cdot 16 = 208$, so $|\psi(w)| \leq 1003$ and $|g| \leq 1003 + \log \pi + \frac{37}{8} < 1100$.

A Chebyshev truncation of degree 447 and the exactness of 224-point Gauss up to degree 447 give the panel error $4\ell H_B 2^{-447}$ for the band entries, and 1100 times this for the gamma-weighted entries. The sum of the panel lengths is 736, and the operator norm is at most 832 times the entrywise bound, so $\epsilon_{\text{band}} = 832 \cdot 4 \cdot 736 \cdot H_B 2^{-447}$ and $\epsilon_K = 1100 \epsilon_{\text{band}}$. For U , the true vectors $E^* h_j$ have norm at most 1, so $\epsilon_U \leq 428(2\epsilon_{\text{band}} + \epsilon_{\text{band}}^2)$; the two parities are orthogonal and the larger coefficient suffices. The pole coefficients use 1704 nodes and Taylor polynomials of degree 1744; with the basis degree at most 1663 the polynomial part is integrated exactly, and the remainder of $e^{\pm x/2}$ is at most $d_p = 2e^{L/2}(L/2)^{1745}/1745!$ uniformly on I . With $\sqrt{2L} < 2$ and $H_p < 3$ this gives a vector error below $4d_p$ and an error in the two rank-one terms below $96d_p + 64d_p^2$. Adding the contributions in ball arithmetic gives $6.01 \times 10^{-97} < 2^{-319}$. $\square$

The constant δ . The saved matrix J was computed with $\delta_{\text{old}} = d(c)$ (enclosed in a ball), while the theorem uses $\delta = 2^{-2901} \leq d(c)$. With $t = \delta_{\text{old}} - \delta \in [0, d(c)]$ we have $h_j^{\text{new}} = h_j^{\text{old}} + t v_j$, so $\|U^{\text{new}} - U^{\text{old}}\| \leq (u_0 + u_1)(2t + t^2) < 2^{-2889}$. This is added to the error of J . The pieces $(I - P)h_j$ do not change, so the projection bounds still apply.

Certificate. As in [9, Section 5], for each parity the proposer outputs rational matrices with denominator 2^{512} : centres A_0, B_0, J_0 , a matrix X , $W_0 \approx \mathcal{W}(X)$, an approximation Y to W_{up}^{-1} , where $W_{\text{up}} = W_0 + \epsilon_W I$, and congruence transforms Z for $C_M = B - mA$, W and T . Put

$$T_0 = Y + J_0 - \alpha I - \beta(B_0 - A_0^2).$$

Since $W_{\text{up}} \succeq \mathcal{W}(X)$, the order of inverses and the source-error bounds give $T \succeq T_0 - \epsilon_T I$, so a congruence test for T_0 gives a lower bound for T . The checker does not import the proposer and computes no eigenvalues. It verifies:

- all entries against the saved source balls, whose SHA-256 sums are pinned, with $\epsilon_A = \epsilon_B = 2^{-400}$ and $\epsilon_J = 2^{-310}$ (quadrature, the δ change and rounding);
- the positivity of $C_M = B - mA$ and of $\mathcal{W}(X)$ by congruence, and $\epsilon_W = 2^{-350}$, $\epsilon_D = 2^{-380}$;
- the inverse residual $r = \|I - W_{\text{up}}Y\| < 1$, giving $\epsilon_Y = \|Y\|r/(1-r)$;
- $\epsilon_T = 2b^2\epsilon_W + \epsilon_Y + \epsilon_J + \beta\epsilon_D \leq 2^{-300}$ and $\eta + \epsilon_T\|Z\|_F^2 < 1$ with $\eta = \|Z^*T_0Z - I\|$, which gives $\lambda_{\min}(T) \geq (1 - \eta - \epsilon_T\|Z\|_F^2)/\|Z\|_F^2$.

The term $2b^2\epsilon_W$ is not needed for the logic, since $W_{\text{up}} \succeq \mathcal{W}(X)$ already accounts for ϵ_W ; it is kept as a margin.

	even	odd
certified upper bound for ϵ_T	4.80×10^{-94}	4.80×10^{-94}
certified lower bound for $\lambda_{\min}(T)$	$2.3979369226 \times 10^{-70}$	$3.9458005898 \times 10^{-66}$

TABLE 2. The certified finite condition (2.2) at $N = 1664$, $\mu = 15$.

The checker passed at 1280 and at 1536 bits, in about 200 seconds and 2.3–2.8 GB of memory each. It rejected seven altered inputs for the intended reasons: the sign of J reversed; an inflated declared error; B replaced by A^2 ; a file changed without updating its hash; an understated error of J ; a wrong contract; and a negative T with the same congruence. An independent reimplementaion of the final check at 1664 bits, which does not share code with the checker, reproduced both lower bounds to 39 digits and rejected an altered entry of J . Independently computed source entries agreed with the saved balls: 18 entries of A , B and J in one review, and 10 further entries in the other, computed by different methods (exact antiderivatives for A and B , 256 Gauss nodes for J); the latter include entries in which omitting the prime 13 changes the value by more than 0.08.

Proof of Theorem O. The certificate gives (2.2) in both parities, hence $G^{-1} + J \succeq \alpha I + \beta D_M$ because $\mathcal{W}(X) \succeq G$. Since $n < m$, the block elimination gives $M + V \succeq 0$ on H . By (2.1), $Q(f) \geq C\delta\|f\|^2 \geq 2^{-2904}\|f\|^2$. $\square$

5. REMARKS

Cost. The source matrices at $N = 1664$ and 3328 bits took about 7800 seconds and 5 GB of memory per parity on a laptop with 16 GB of memory, which is close to the limit of what we can do locally. Larger windows need a more economical implementation.

Margins. The certified finite margins are about 2^{-232} (even) and 2^{-218} (odd), while all paid errors are below 2^{-300} . The final constant 2^{-2904} is much smaller because it comes from the out-of-band bound, which is of size $e^{-2\Omega L}$.

Trust. The proof relies on the analytic lemmas above and in [8, 9], on Arb/FLINT [1] for the ball enclosures, and on the checker program. It has not been checked by a proof assistant.

REPRODUCIBILITY

The proposer, the checker, the generator, the independent check and the mutation tests are in the folder `certificate-VIII/` of release v1.6.0 of <https://github.com/moriryota/prolate-weil-bounds>, archived on Zenodo, doi:10.5281/zenodo.23092422 (all versions). The certificate matrices and the source matrices (about 1.8 GB) are in the separate repository <https://github.com/moriryota/weil-log15-certificate-data>, archived on Zenodo, doi:10.5281/zenodo.23281176, with a script that restores them into the code folder and checks every SHA-256 sum. The checker runs in about 200 seconds and 3 GB of memory.

USE OF GENERATIVE AI

This work was carried out with extensive use of generative AI systems, namely Anthropic Claude and OpenAI GPT. Under the direction of the author, these systems derived the analytic bounds, implemented and ran the computations, and reviewed one another’s work adversarially. The author is responsible for the content.

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